

Semyon Davshits

Data Scientist / ML Engineer · Internship

Moscow, Russia · +7 977 284-01-54 · s.davshits@yandex.ru · t.me/SeMe4KO0 · github.com/SeMe4K-0

Fourth-year student at Bauman Moscow State Technical University, Dept. IU5. Models on tabular data and time series, uplift modelling and experiment statistics; also computer vision and audio. I take models through to a service with an API and Docker.

PROJECTS

X5 RetailHero - uplift modelling: who actually changes behaviour when contacted

CatBoost · PostgreSQL · dbt

- T-, S- and X-learners on CatBoost over properly randomised data; Qini, AUUC and uplift@k implemented from scratch. Features from 45.8M receipt lines with point-in-time correctness (PostgreSQL, dbt, 42 tests).
- Ranking by response probability is worse than random: the top deciles by response and by uplift overlap by only **0.44%** against 10% expected by chance. The uplift model gives $\Delta \text{uplift@10\%} = +9.53 \text{ pp}$ [+6.63; +12.54], 2,000-replicate bootstrap.
- Response-based targeting has a statistically detectable effect yet loses money: 0.53 pp against a 4.67 pp break-even. Switching to uplift targeting is worth **₽612k** per campaign.
- The analysis plan was pre-registered before any model was trained; a mutation test proves the leakage auditor catches look-ahead bias.

E-CUP 2026 (Ozon) - 30-day GMV forecast per user

14th of 316 · team of 2

- 250,000 users and 409 days of search and catalogue history. Every training row is a user-date pair with features computed strictly before that date, so training mirrors inference.
- Adversarial validation exposed an anchor-date leak (AUC 1.0); removing it showed that 93% of the local improvement had been illusory.
- Trained directly in the scale of the metric (log1p) with an ensemble of a GRU and gradient boosting. **RMSLE 1.6631** on the private leaderboard.

OreScope - ore grade classification from thin-section microscopy

Nornickel AI Science Hack, finalist

- U-Net phase segmentation on microscopy panoramas of up to 300 MP; the grade follows from an explicit expert rule over the mask, with a confidence map and a written verdict.
- Camera domain shift: colour normalisation lifted detection on the new domain from 5 to 14 samples out of 20 without a single new label.
- **macro-F1 0.91** on intergrowth type, **0.77** end-to-end. A panorama takes 124 s against a 5-minute requirement; FastAPI, Docker, built in 48 hours.

Stem Separator - splitting a track into four instrumental stems

Demucs · FastAPI · Docker

- Separates vocals, drums, bass and other with Demucs, measured with museval on MUSDB18-sample (5 tracks): **SDR 9.34 dB** on vocals and **9.90 dB** on bass.
- A three-minute track in 14 seconds. REST API, web interface and CLI, automatic MPS/CUDA/CPU selection, 19 tests, Docker.

RESEARCH

Evaluating generative music models with Fréchet Audio Distance

Bauman MSTU, Dept. IU5

- Built a benchmark of five open-source text-to-music models across FAD, FAD-inf and CLAP Score, with three embedding models and two reference datasets.
- Showed that the ranking flips with the reference set: **AudioLDM-M** leads on FMA-Pop and **MusicLDM** on MTG-Jamendo, so a FAD figure means little across papers unless the reference is stated.

ACHIEVEMENTS

- [Yandex ML Challenge 2026](#) - **42nd of 100 finalists**.
- [Yandex School of Data Analysis](#), "AI Agents Security Week" intensive. Classifier for LLM requests under a Trust & Safety policy: 4 actions and 13 reason codes, a deterministic TF-IDF matcher. Raised the starter solution from 61.54 to **99.81 out of 100**. Red-teaming system-prompt extraction - **71.43 out of 100**.
- [DatsSol hackathon 2026](#) - **16th of 166**.
- [Nornickel AI Science Hack 2026](#) - finalist.

EDUCATION

Bauman Moscow State Technical University, Faculty of Informatics and Control Systems, Dept. IU5 - BSc, 4th year, expected 2027
Information Processing and Control Systems

Relevant coursework: machine learning, probability and mathematical statistics, databases, operations research.

SKILLS

Core: Python, pandas, polars, scikit-learn, CatBoost, LightGBM, PyTorch

Data: SQL, PostgreSQL, window functions, dbt

Statistics: hypothesis testing, bootstrap, multiple-comparison correction, power and MDE, uplift modelling

ML and DL: tabular data and time series, segmentation and detection, audio

Engineering: Docker, FastAPI, pytest, Git, Linux

Languages: Russian (native), English (B2)